

Little Red Dots: Super-Eddington Wind-Shrouded Engines

Type IIIn supernovae and eta Carinae's Great Eruption offer a unified model for LRDs.

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Background

JWST's Little Red Dots show blackbody-like continua, broad Balmer lines, and Balmer breaks. Similar features appear in Type IIIn supernovae and eta Carinae's Great Eruption.

LRDs' overmassive black hole inferences may misinterpret wind physics as a virial broad-line region.

What this paper is selling

The authors propose that dense, slow winds create a pseudo-photosphere hiding an intermediate-mass engine. Fast winds shock against this envelope, powering the emission and explaining the full LRD lifecycle.

Key results to remember

- Escape velocity limits engine mass to below 10^5 solar masses.
- LRDs are intermediate-mass (10^3 - 6 solar masses), super-Eddington ($L/L_{\text{edd}} > 5$) systems.
- Dust in the envelope may seed classical AGN.

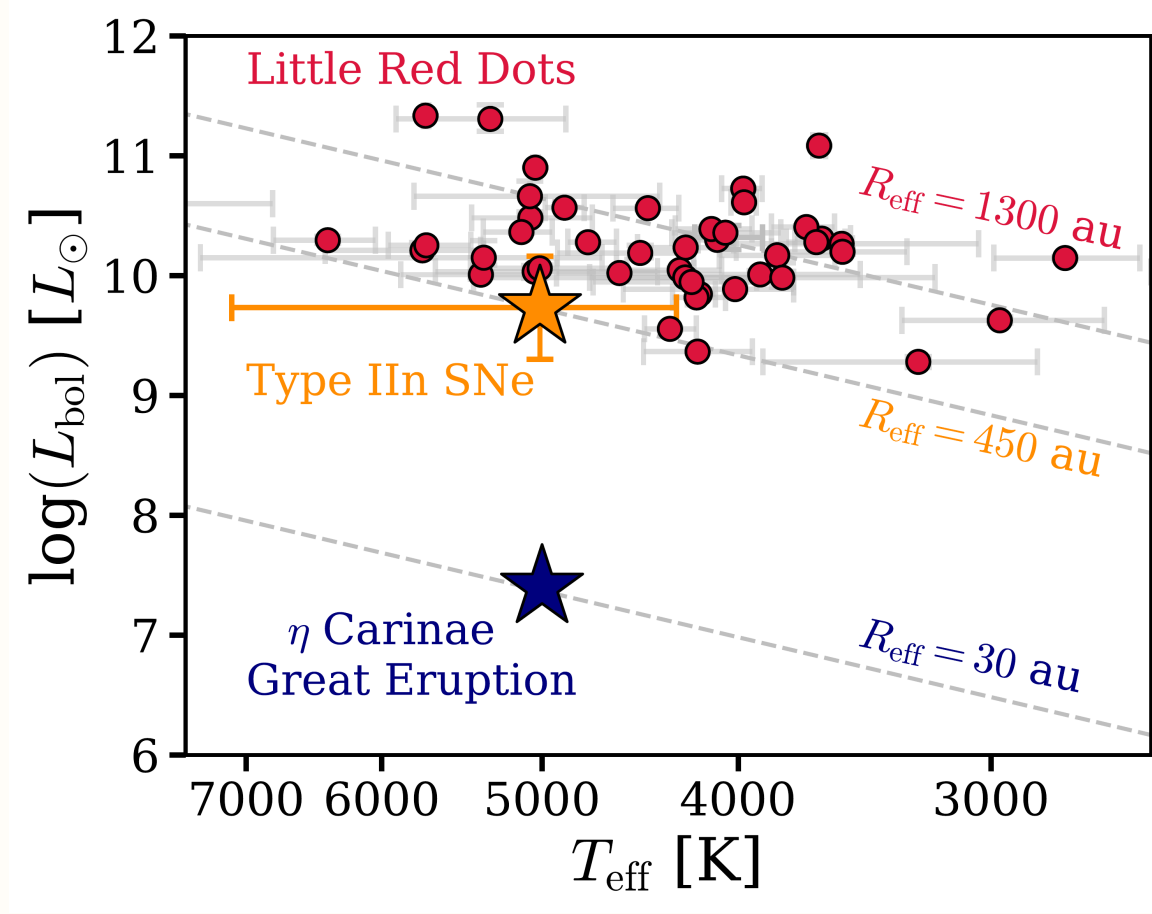


Figure 4 | LRDs, Type IIIn SNe, and the Great Eruption cluster at similar effective temperatures despite spanning four decades in luminosity.

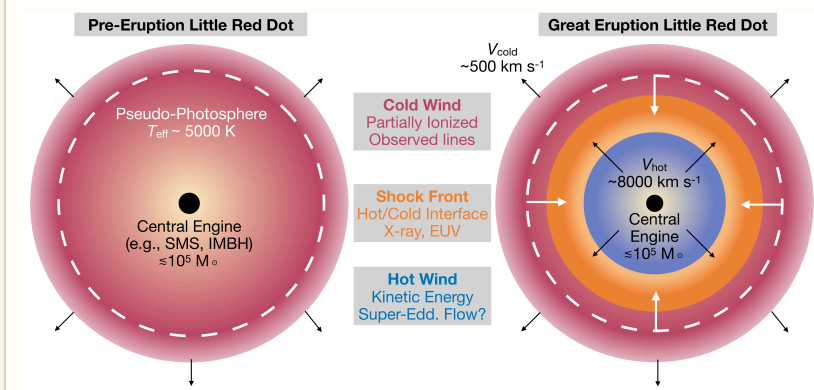


Figure 8 | Schematic shows a cold wind creating a pseudo-photosphere, then hot winds shocking to power the LRD.